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SUPERSEDING
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FEDERAL SPECIFICATION

TOWELS, PAPER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers folded and roll towels of the wet-strength type intended for use as disposable wipers for drying the hands.

1.2 Classification. The paper towels shall be of the following types and sizes as specified (see 6.1):

Type I - Singlefold.
Type II - Multifold.
  Size 1.
  Size 2.
Type III - C-fold.
Type IV - Roll towels.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

PPP-B-636 - Boxes, Shipping, Fiberboard.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Technical Association of the Pulp and Paper Industry (TAPPI) Standards:

T 404 - Tensile Breaking Strength of Paper and Paperboard.
T 410 - Weight per Unit Area (Basis Weight or Substance) of Paper and Paperboard.
T 432 - Water Absorption of Bibulous Paper.
T 456 - Wet Tensile Breaking Strength of Paper and Paperboard.

(Application for copies should be addressed to the Technical Association of the Pulp and Paper Industry, One Dunwoody Park, Atlanta, GA 30341.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. The towels shall be made from fibrous material which shall conform to the requirements in table I when tested as specified in 4.4. The material offered shall contain not less than 95 percent of recovered materials. Recovered materials are defined as material which has been collected or recovered from solid waste. Solid waste is defined as any garbage, refuse, sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility and other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities, but does not include solid or dissolved material in domestic sewage, or solid or dissolved materials in irrigation return flows or industrial discharges.

TABLE I. Physical characteristics

Characteristic	All types and sizes
Basis weight (24 x 36 inches - 500 sheets)	Minimum-25 pounds. Maximum-30 pounds.
Water absorption of 0.1 ml water	Maximum-100 seconds.
Tensile strength	
Dry	Minimum-1.0 kg/15 mm
Wet, after accelerated aging (see 4.5.2)	Minimum-0.275 kg/15 mm

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3.2 Construction.

3.2.1 Sheets. Type I and II towels shall be interlocked to facilitate dispensing. Types I, II and III shall fit and feed satisfactorily from the dispensing cabinet specified in the invitation to bid.

3.2.2 Rolls. Type IV towels shall be in rolls of 150 with each towel perforated completely and uniformly across its width so that the towel can be torn from the roll at the perforations only. The perforations shall be so spaced that the individual towels conform to the minimum width and area specified in table II.

3.3 Size and area. Towels of the type and size specified shall meet the minimum area and minimum smaller dimension requirements indicated in table II.

TABLE II. Sizes and areas

Types	Shortest dimension 1/ Minimum (inches)	Minimum area (square inches)
I	9-1/8	100
II		
Size 1	9-1/8	85
Size 2	9-1/8	130
III	10-1/8	140
IV	11	140

1/ The shorter side of the unfolded sheet.

3.4 Surface finish and formation. The towels shall have a normal, commercial surface finish. Formation shall be uniform, with no clumps of undisbursed fibers.

3.5 Odor. The towels shall have no disagreeable odor, when tested in accordance with 4.5.1.

3.6 Standard comparison sample. Paper towels shall compare favorably with the standard comparison sample with respect to surface finish, formation, cleanliness, and workmanship (see 6.2).

3.7 Workmanship and cleanliness. Paper towels shall be free of holes, tears, and wrinkles. Edges shall be cleanly cut and not ragged. There shall be no evidence of foreign materials or dirt.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

4.2 Certification and sampling.

4.2.1 Certification. A certificate shall be submitted with each bid specifying that the material to be supplied will contain the percentage of recovered material required herein.

4.2.2 Sampling. Sampling for inspection and testing shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.3 End item examination.

4.3.1 Examination of preparation for delivery. An examination shall be made for defects in unit load size and weight, marking, materials, workmanship and contents (quantity) to determine conformance with the requirements of section 5. The inspection level shall be S-2 with an AQL of 4.0 percent defective. The sample unit shall be one shipping container or unit load, as applicable.

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4.3.2 Sheet and roll examination. Examine the towels in unit packages for the defects shown below. The sample unit shall be one package or roll. The inspection level and AQL shall be S-3 and 4.0 percent defective.

Examine	Defects
Sheets	Not type specified; not interlocked, as applicable.
Rolls	Not completely perforated; not perforated along entire width of sheet. Perforations not perpendicular to edges of sheet. Perforations wavy, not uniform. Sheets separated at perforations.

4.3.3 Examination of the item for count. Two packages or rolls shall be selected at random from each of the shipping containers selected in 4.3.1. The lot shall be unacceptable if the average count of towels per package or sheets per roll, of all packages examined, is less than specified.

4.3.4 Examination for visual and dimensional defects. Examine the towels for the defects shown below. An inspection level of S-3 and an AQL of 4.0 percent defective shall be used. The sample unit is one towel.

Examine	Defects
Appearance	Formation, surface finish, and cleanliness does not compare favorably with comparison sample.
	Any uneven or ragged edges, holes, tears or wrinkles.
Shortest dimension	Less than specified.
Area	Less than specified.

4.4 End item testing. Each lot of towels shall be tested for the characteristics in table III. The sample unit shall be one package or roll. The inspection level shall be S-1 with an AQL of 4.0 percent defective. The requirements of table I shall be applied to the values obtained for each sample unit. The tensile strength results shall be calculated as the average in both directions (combined).

TABLE III. Instructions for testing

Characteristic	Specification reference	
	Requirement	Test method
Basis weight	Table I	T 410 1/
Water absorption	Table I	T 432
Tensile strength		
Dry	Table I	T 404
Wet	Table I	T 456 and 4.5.2
Odor	3.5	4.5.1

1/ TAPPI test method.

4.5 Test methods.

4.5.1 Test for odor. Soak one towel or sheet, as applicable, in distilled water at 73°F for two minutes. Then remove the towel from the water and smell it from a distance of 24 inches. Any disagreeable odor shall be grounds for rejection.

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4.5.2 Wet tensile test. Before testing for wet tensile strength according to T 456, the samples of towels shall be placed in a 105°C oven for 10-12 minutes. The samples shall then be conditioned for at least 4 hours at 50 ± 2% RH and 23 ± 1°C (73.4 ± 1.8°F) before testing.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The towels shall be preserved and packaged in accordance with normal commercial practice. The complete package shall be designed to protect the item against damage during shipment, handling and storage.

5.2 Packing. Packing shall be level A, or Commercial as specified.

5.2.1 Level A. Towels packaged as specified in 5.1 shall be packed in a box conforming to PPP-B-636, class weather-resistant. The box shall be closed, waterproof sealed, and banded in accordance with the appendix to the box specification.

5.2.2 Commercial. Towels packaged as specified in 5.1 shall be packed in a fiberboard box in accordance with normal commercial practice. The complete pack shall be designed to protect the item against damage during shipment, handling and storage, to insure safe delivery at destination, to provide for safe redistribution by the initial receiving activity and be acceptable by common carrier under National Motor Freight Classification or Uniform Freight Classification.

5.3 Unitization. When shipments to Government depots are full car or truckload, the shipping containers shall be unitized for shipment and handling in accordance with normal commercial practice. The unitized load shall not exceed 2,500 pounds in weight, 63 inches in height, 56 inches in length and 45 inches in width.

5.4 Marking. Shipping containers and unitized loads (when applicable) shall be marked as specified in the contract or purchase order (see 6.1).

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and size required (see 1.2 and table II).
(c) Type of dispensing cabinet to be used (see 3.2.1).
(d) Packaging and packing required (see 5.1 and 5.2).
(e) Marking required (see 5.4).

6.2 Standard comparison samples may be obtained upon application to the procuring agency issuing the invitation for bids.

6.3 Department of Defense has waived coordination of this specification.

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 60 cents each.

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